

Customer Behavior Analysis

Data Analyst Project Report

1. Report Structure & Executive Summary

This project demonstrates an end-to-end Data Analyst workflow using a customer shopping behavior dataset. The analysis starts with raw CSV data, applies data cleaning and feature engineering in Python, answers business questions using SQL Server, and delivers the findings through an interactive Power BI dashboard.

Project metric	Value
Records analyzed	3,900
Unique customers	3,900
Total revenue	233,081.00
Average purchase amount	59.76
Average review rating	3.75

Executive Summary

- Cleaned and standardized the customer dataset using Python/Pandas.
- Handled missing review ratings using category-level median imputation.
- Created age groups and purchase-frequency features for downstream analysis.
- Used SQL aggregation, subqueries, CASE statements, CTEs, and window functions to answer business questions.
- Built a Power BI dashboard with KPIs, slicers, category-level sales/revenue visuals, subscription mix, and age-group analysis.
- Converted transactional data into insights relevant to customer retention, discount strategy, product performance, and revenue management.

2. Data Pipeline & Methodology

The pipeline follows a structured progression from raw data to business-ready insight.

Stage	Tool	Activity
1. Ingestion	Python / Pandas	Read the raw CSV dataset.
2. Cleaning	Python / Pandas	Standardize column names, handle missing values and remove redundant fields.
3. Feature engineering	Python / Pandas	Create age groups and purchase-frequency days.
4. Export	CSV	Save the cleaned analytical dataset.
5. Analysis	SQL Server	Run business questions using aggregations, CTEs and window functions.
6. Visualization	Power BI	Build an interactive dashboard.
7. Delivery	PDF report	Document methodology, outputs and business value.

Python Cleaning & Feature Engineering

```
df = pd.read_csv('/content/customer_shopping_behavior.csv')

df.columns = df.columns.str.lower()
df.columns = df.columns.str.replace(' ', '_')

df['review_rating'] = df['review_rating'].fillna(
    df.groupby('category')['review_rating'].transform('median')
)

labels = ['younger_adult', 'adult', 'middle_age', 'senior']
df['age_group'] = pd.qcut(df['age'], q=4, labels=labels)

df = df.drop(columns=['promo_code_used'], axis=1)
df.to_csv('customer_data_cleaned.csv', index=False)
```

The cleaning process standardizes the schema, fills missing review ratings with the median of the customer's category, creates quartile-based age groups, removes a redundant promotion field, and exports a clean dataset for SQL analysis.

3. Analytical Techniques & Frameworks

The analysis combines descriptive analytics with SQL techniques commonly used in real-world Data Analyst work.

Technique	Use in project
GROUP BY	Compare revenue, average spend and counts across business dimensions.
SUM / AVG / COUNT	Measure revenue, purchase value, ratings and customer counts.
Subqueries	Compare individual purchases with the overall average.
CASE	Segment customers based on previous purchases.
CTE	Break complex analysis into readable intermediate steps.
ROW_NUMBER()	Rank products within each category.
Conditional aggregation	Calculate discount utilization percentages.
Pandas groupby / transform	Category-level missing-value treatment.
pd.qcut	Create four age groups based on quartiles.

Representative Window-Function Pattern

```
WITH product_orders AS (  
  SELECT category, item_purchased, COUNT(*) AS total_orders  
  FROM customer_data_cleaned  
  GROUP BY category, item_purchased  
)  
ranked_products AS (  
  SELECT *,  
  ROW_NUMBER() OVER (  
    PARTITION BY category  
    ORDER BY total_orders DESC  
  ) AS product_rank  
  FROM product_orders  
)  
SELECT *  
FROM ranked_products  
WHERE product_rank <= 3;
```

This framework first calculates purchase volume for each product within a category, then ranks the products separately inside each category. It is more appropriate than ranking customer IDs when the business question asks for the most purchased products.

4. Tooling, Visualization & Delivery

Tool	Role	Deliverable
Python / Pandas	Data preparation	Clean dataset and engineered features
SQL Server	Business analysis	Queries, segmentation, ranking and KPIs
Power BI	Visualization	Interactive dashboard
PDF	Communication	Portfolio-ready project report

Dashboard Delivery

- KPI cards communicate customer count, average purchase amount and average review rating.
- Slicers provide interactive filtering by subscription status, gender and category.
- Category charts show revenue and sales contribution.
- Age-group visuals support demographic revenue and sales comparisons.
- The dashboard acts as the final presentation layer for the SQL and Python analysis.

5. Business Value & Outcomes

The analysis is designed to support practical decisions across customer retention, marketing, product management, discount strategy and revenue planning.

Business area	Outcome / decision support
Customer retention	Identify New, Returning and Loyal customers for targeted engagement.
Subscription strategy	Compare subscriber and non-subscriber average spend and total revenue.
Discount strategy	Find products with high discount utilization and evaluate high-value discounted purchases.
Product performance	Identify highly rated products and the most purchased products within each category.
Revenue management	Compare revenue across gender, category and age groups.
Operations	Compare average purchase amounts for Standard and Express shipping.
Management reporting	Provide a single interactive Power BI view for business exploration.

Overall Analyst Outcome

The project demonstrates the complete analytics lifecycle: data preparation in Python, structured analysis in SQL, interactive visualization in Power BI, and communication of business value through a documented report. This makes the project suitable as a Data Analyst portfolio case study.

6. Business Questions, SQL Analysis & Output

The following sections document the ten business questions analyzed in the project. For each question, the business objective is described first, followed by the SQL used and a small screenshot of the resulting output.

Q1. What is the total revenue generated by male vs. female customers?

Description: Compares total purchase revenue across genders to understand demographic contribution to revenue.

SQL:

```
SELECT gender,
SUM(purchase_amount) AS total_revenue
FROM customer_data_cleaned
GROUP BY gender;
```

Output screenshot:

Output snapshot — revenue by gender

gender	total_revenue
Male	157,890
Female	75,191

Q2. Which customers used a discount but still spent more than the average purchase amount?

Description: Identifies high-value purchases where a discount was applied and the purchase amount was above the dataset-wide average.

SQL:

```
SELECT customer_id, purchase_amount
FROM customer_data_cleaned
WHERE discount_applied = 'Yes'
AND purchase_amount > (
SELECT AVG(purchase_amount)
FROM customer_data_cleaned
);
```

Output screenshot:

Output snapshot — first 10 matching customers (overall average = 59.76)

customer_id	purchase_amount
582	100
616	100
519	100
194	100
1,480	100
244	100
205	100
249	100
1,422	100
770	100

Q3. Which are the top 5 products with the highest average review rating?

Description: Ranks products by their average customer review rating and returns the five highest-rated products.

SQL:

```
SELECT TOP 5
item_purchased,
ROUND(AVG(review_rating), 2) AS average_rating
FROM customer_data_cleaned
GROUP BY item_purchased
ORDER BY average_rating DESC;
```

Output screenshot:

Output snapshot — top 5 products by average rating

item_purchased	average_rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Q4. Compare the average purchase amounts between Standard and Express Shipping.

Description: Measures whether average order value differs between the two shipping methods.

SQL:

```
SELECT shipping_type,
AVG(purchase_amount) AS average_purchase
FROM customer_data_cleaned
WHERE shipping_type IN ('Express', 'Standard')
GROUP BY shipping_type;
```

Output screenshot:

Output snapshot — average purchase by shipping type

shipping_type	average_purchase
Express	60.48
Standard	58.46

Q5. Do subscribed customers spend more?

Description: Compares average spend and total revenue between subscribers and non-subscribers.

SQL:

```
SELECT subscription_status,
AVG(purchase_amount) AS average_spend,
SUM(purchase_amount) AS revenue
FROM customer_data_cleaned
GROUP BY subscription_status;
```

Output screenshot:

Output snapshot — subscriber vs non-subscriber

subscription_status	average_spend	revenue	customers
No	59.87	170,436	2,847
Yes	59.49	62,645	1,053

Q6. Which 5 products have the highest percentage of purchases with discounts applied?

Description: Calculates discount utilization by product using conditional aggregation and identifies the top five.

SQL:

```
SELECT TOP 5
item_purchased,
ROUND(
100.0 * SUM(
CASE WHEN discount_applied = 'Yes'
THEN 1 ELSE 0 END
) / COUNT(*), 2
) AS discount_rate
FROM customer_data_cleaned
GROUP BY item_purchased
ORDER BY discount_rate DESC;
```

Output screenshot:

Output snapshot — top 5 discount rates

item_purchased	discounted_purchases	total_purchases	discount_rate
Hat	77	154	50.00
Sneakers	72	145	49.66
Coat	79	161	49.07
Sweater	79	164	48.17
Pants	81	171	47.37

Q7. Segment customers into New, Returning, and Loyal based on previous purchases.

Description: Uses CASE logic to convert purchase history into customer segments and then counts customers in each segment.

SQL:

```
WITH customer_segments AS (  
  SELECT customer_id,  
  CASE  
    WHEN previous_purchases = 1 THEN 'New'  
    WHEN previous_purchases BETWEEN 2 AND 10 THEN 'Returning'  
    ELSE 'Loyal'  
  END AS customer_segment  
  FROM customer_data_cleaned  
)  
SELECT customer_segment,  
COUNT(DISTINCT customer_id) AS customer_count  
FROM customer_segments  
GROUP BY customer_segment  
ORDER BY customer_count DESC;
```

Output screenshot:

Output snapshot — customer segment counts

customer_segment	customer_count
New	83
Returning	701
Loyal	3,116

Q8. What are the top 3 most purchased products within each category?

Description: Ranks products by purchase volume independently within each category. This is a window-function use case.

SQL:

```
WITH product_orders AS (  
  SELECT category,  
  item_purchased,  
  COUNT(*) AS total_orders  
  FROM customer_data_cleaned  
  GROUP BY category, item_purchased  
)  
, ranked_products AS (  
  SELECT *,  
  ROW_NUMBER() OVER (  
    PARTITION BY category  
    ORDER BY total_orders DESC  
  ) AS product_rank  
  FROM product_orders  
)  
SELECT category,  
item_purchased,  
total_orders,  
product_rank  
FROM ranked_products  
WHERE product_rank <= 3;
```

Output screenshot:

Output snapshot — top 3 products within each category

category	item_purchased	total_orders	rank
Accessories	Jewelry	171	1
Accessories	Belt	161	2
Accessories	Sunglasses	161	3
Clothing	Blouse	171	1
Clothing	Pants	171	2
Clothing	Shirt	169	3
Footwear	Sandals	160	1
Footwear	Shoes	150	2
Footwear	Sneakers	145	3
Outerwear	Jacket	163	1
Outerwear	Coat	161	2

Q9. Are repeat buyers (more than 5 previous purchases) also likely to subscribe?

Description: Counts repeat buyers by subscription status to compare subscription behavior among customers with more than five previous purchases.

SQL:

```
SELECT subscription_status,  
COUNT(DISTINCT customer_id) AS repeat_buyers  
FROM customer_data_cleaned  
WHERE previous_purchases > 5  
GROUP BY subscription_status;
```

Output screenshot:

Output snapshot — repeat buyers by subscription status

subscription_status	repeat_buyers	share_pct
No	2,518	72.44
Yes	958	27.56

Q10. What is the revenue contribution of each age group?

Description: Aggregates purchase revenue by the quartile-based age groups created during Python feature engineering.

SQL:

```
SELECT age_group,  
SUM(purchase_amount) AS total_revenue  
FROM customer_data_cleaned  
GROUP BY age_group  
ORDER BY total_revenue DESC;
```

Output screenshot:

Output snapshot — revenue by age group

age_group	total_revenue	revenue_share_pct
younger_adult	62,143	26.66
adult	55,978	24.02
middle_age	59,197	25.40
senior	55,763	23.92

7. Final Power BI Dashboard

The following image is the final dashboard delivered from the analysis. It is intentionally placed at the very end of the report as the final visual summary of the project.

